AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

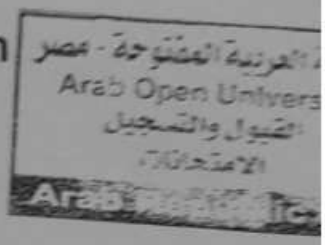
FIN340-F

Course Title: ASSET PRICING AND CORPORATE FINANCE

Version B

Mid-term Examination
2025-2026 / Spring

15 – April – 2026



Number of Exam Pages: 7 (including this cover sheet)	Time Allowed: 1.5 hours

Instructions:

Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
This is a **CLOSED** book examination. NO documents of any kind are allowed.
You **MUST** answer the questions using only the answer booklets provided
You **MUST** return the examination paper together with your answer booklets.
Do **NOT** write your answers in pencil except graphs/sketches
Students are **NOT** allowed to use any type of dictionary or mobile phones during the ex
Students are allowed to use ***non-programmable*** calculators only (if needed).
Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Pol

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Solve All Questions

- **Question 1: (25 marks)**

You have just made a $1,500 contribution to your individual retirement account. Assume you earn a 12 percent rate of return and make no additional contributions. How much more will your account be worth when you retire in 25 years than it would be if you waited another 10 years before making this contribution?

- **Question 2: (30 marks)**

Given the below, what is the operating cash flow for 2025?

Cost of goods sold= $6,550

Interest= 450

Dividends= 600

Depreciation= 960

Change in retained earnings= 810

Tax rate= 34%

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- **Question 3: (30 marks)**

Bale Inc. had equity of $375,000 at the beginning of the year. At the end of the year, the company had total assets of $300,000. During the year, the company sold no new equity. Net income for the year was $80,000 and dividends were $50,000. What is the sustainable growth rate?

dividends 50 000
net income 80 000
Total assets 300 000
equity 375 000

- **Question 4: (15 marks)**

Define and explain the Agency problem.

End of Questions

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Formulas

Cash Flow Summary

I. The cash flow identity

Cash flow from assets	=	Cash flow to creditors (bondholders)
	+	stockholders (owners)

II. Cash flow from assets

Cash flow from assets	=	Operating cash flow
	−	Net capital spending
	−	Change in net working capital (NWC)

where:

Operating cash flow	=	Earnings before interest and taxes (EBIT)
	+	Depreciation − Taxes
Net capital spending	=	Ending net fixed assets − Beginning net fixed assets
	+	Depreciation
Change in NWC	=	Ending NWC − Beginning NWC

III. Cash flow to creditors (bondholders)

Cash flow to creditors	=	Interest paid − Net new borrowing

IV. Cash flow to stockholders (owners)

Cash flow to stockholders	=	Dividends paid − Net new equity raised

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Common Financial Ratios

I. Short-term solvency, or liquidity, ratios
$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$
$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$
$\text{Cash ratio} = \frac{\text{Cash}}{\text{Current liabilities}}$
$\text{Net working capital to total assets} = \frac{\text{Net working capital}}{\text{Total assets}}$
$\text{Interval measure} = \frac{\text{Current assets}}{\text{Average daily operating costs}}$
II. Long-term solvency, or financial leverage, ratios
$\text{Total debt ratio} = \frac{\text{Total assets} - \text{Total equity}}{\text{Total assets}}$
$\text{Debt-equity ratio} = \text{Total debt/Total equity}$
$\text{Equity multiplier} = \text{Total assets/Total equity}$
$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Total equity}}$
$\text{Times interest earned ratio} = \frac{\text{EBIT}}{\text{Interest}}$
$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest}}$

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V. Market value ratios

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

$$\text{PEG ratio} = \frac{\text{Price-earnings ratio}}{\text{Earnings growth rate (\%)}}$$

$$\text{Price-sales ratio} = \frac{\text{Price per share}}{\text{Sales per share}}$$

$$\text{Market-to-book-ratio} = \frac{\text{Market value per share}}{\text{Book value per share}}$$

$$\text{Tobin's Q ratio} = \frac{\text{Market value of assets}}{\text{Replacement cost of assets}}$$

$$\text{Enterprise value-EBITDA ratio} = \frac{\text{Enterprise value}}{\text{EBITDA}}$$

ROE= Profit Margin X Total Asset Turnover X Equity Multiplier

Summary of Internal and Sustainable Growth Rates

I. Internal Growth Rate

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}$$

where:

ROA = Return on assets= Net income/Total assets
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The internal growth rate is the maximum growth rate that can be achieved with no external financing of any kind.

II. Sustainable Growth Rate

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b}$$

where:

ROE = Return on equity= Net income/Total equity
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The sustainable growth rate is the maximum growth rate that can be achieved with no external equity financing while maintaining a constant debt-equity ratio.

Dividend payout ratio = Cash dividends/Net income

Retention ratio= 1- the dividend payout ratio

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$$EPS = \frac{\text{Net income}}{\text{Shares outstanding}} =$$

Net Working Capital= Current Asset- Current Liabilities

$$PV = FV/ (1 + r)^t$$

$$FV = PV^* (1 + r)^t$$

$$r = (FV / PV)^{1/t} - 1$$

$$t = \ln(FV / PV) / \ln(1 + r)$$

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